

INTERIM REPORT

WHO INFLUENZA CENTRE

LONDON

MARCH 2006

**CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2 AND B
VIRUSES ISOLATED OCTOBER 2005 TO FEBRUARY 2006**

**WHO Collaborating Centre for
Reference and Research on Influenza
National Institute for Medical Research
The Ridgeway
Mill Hill
London, NW7 1AA**

**Dr. Alan J. Hay (Director)
Dr. Yi Pu Lin (Assistant Director)
Miss Victoria Gregory
Mr. Michael Bennett**

Tel: 020 8959 3666; 020 8816 2152

Fax: 020 8906 4477

Email: ahay@nimr.mrc.ac.uk; lyipu@nimr.mrc.ac.uk

Of some 200 influenza viruses isolated in 19 countries since October 2005, approximately half (49%) were A H1N1 and 16% were A H3N2 (Table 1). Influenza B viruses accounted for 35% of isolates, 75% of which were of the B/Victoria lineage and 25% were of the B/Yamagata lineage.

AH1N1 viruses

The majority of H1N1 viruses were antigenically closely related to the current vaccine strain A/New Caledonia/20/99 and to the more recent reference viruses A/Netherlands/128/2004 (MDCK), A/Hong Kong/2367/04 (MDCK) and A/Thessaloniki/24/2005 (egg) (Table 2).

The HA sequences of recent isolates fell into two phylogenetic groups (Figure 1): 1) represented by A/Hong Kong/2367/2004 and characterised by the amino acid changes T82K, Y94H, R145K, R208K and T266N (red in Figure 1), and prominent among Asian isolates; 2) represented by A/Thessaloniki/24/2005 and characterized by the changes Y252F and V314A (blue in Figure 1) (Table 3). There was relatively little change in the sequences of these variants over the past 12-18 months; however, the proportions of the former increased to a level similar to that of the latter. No consistent differences in HI were observed between variants of the two phylogenetic groups. The NA sequences of a number of the latter viruses were also shown to be similar to those of earlier isolates (Figure 2, Table 3); however, others, represented by A/England/488/2005, possessed the amino acid changes A16T, S59N, I117T, V234M, N273S, T339N and D383N and were more closely related to the NAs of viruses with the former group 1 HAs.

AH3N2 viruses

Of the relatively few H3N2 viruses received, the majority were antigenically closely related to the recent MDCK-grown reference viruses A/Hong Kong/4443/2005 and/or A/Annecy/1138/2005. In general, they gave low HI titres, relative to the homologous virus, with egg-grown vaccine strains, A/California/7/2004 and A/New York/55/2004, and the recently recommended strain A/Wisconsin/67/2005. These differences are due in large part to the cells in which the viruses are grown.

The majority of HA sequences of recent isolates were characterized by the changes S193F and D225N and represented by A/Hong Kong/4443/2005 and A/Wisconsin/67/05 (red in Figure 3). A number fell within the older phylogenetic group characterized by amino acid changes V112I, N145S and K173E, with additional changes D188Y, G175V or S199P (blue in Figure 3).

M2 protein sequences obtained for viruses with HAs within the former group all possessed N31, indicative of amantadine resistance.

Influenza B viruses

The majority of **B/Yamagata-lineage** (HA) viruses were antigenically closely related to the recent egg-grown reference strains, B/Egypt/144/2005 and B/Florida/7/2004, and gave reduced HI titres with the vaccine strains B/Shanghai/361/2002 and

B/Jiangsu/10/2003 (Table 6). Some were antigenically more closely related to B/Jiangsu/10/03.

Most of the HA sequences of recent isolates were similar to those of viruses circulating during 2005 and were characterized by the change V252M, many with an additional change P108A, relative to the HA sequence of B/Shanghai/361/2002 (green in Figure 5, Table 8). A number of viruses from Norway had the additional changes N166Y and G230D.

Most of the **B/Victoria-lineage** (HA) viruses were antigenically closely-related to the MDCK-grown reference strain B/Hong Kong/45/2005 (Table 7). However, many gave reduced HI titres with the egg-grown vaccine strain B/Malaysia/2506/2005.

Most of the HA sequences of recent isolates fell into one of two principal phylogenetic groups (Figure 6): 1) characterized by the amino acid changes R48E, K80R and K129N relative to B/Shandong/7/97, and represented by B/Hong Kong/45/2005 and B/Malaysia/2506/2004 (red in Figure 6, Table 8); 2) characterized by the changes, K80E plus V190I and represented by B/Ohio/1/05 (blue in Figure 6). Some sequences have additional changes N75K and V177I, such as B/Toulouse/44/06. We did not observe any obvious differences in HI tests between viruses belonging to the different phylogenetic groups.

Acknowledgements

In addition to data received from the other 3 Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Denmark, Germany, Italy, Norway, Spain, Sweden and the U.K. (indicated in Figures by suffixes).

Table 1: Human influenza isolates, October 2005- February 2006

Country	H1N1	H3N2	B	
			Yamagata lineage	Victoria lineage
October				
Sweden		1		
Switzerland	1			
November				
France	1			
Germany	1			
Hong Kong	3			
Madagascar	4	1		
Sweden		2		
UK	1			
December				
Denmark		1		
Egypt	4			
France	6	1	4	6
Hong Kong	4	2		6
Ireland		1		
Madagascar	22			
Netherlands		1		
Norway	2	2	7	2
Portugal	1			
Sweden		2		
UK	3	1		5
January				
Denmark			1	
Egypt	16			4
Finland		1		
France	1	1	1	
Hong Kong	5			
Iceland		2		
Ireland				1
Italy			1	1
Latvia		2		3
Netherlands		1		2
Norway	3	2	3	5
Portugal	7			3
Slovakia				2
Spain	4			3
Sweden	2	2	1	1
Switzerland	1			1
UK		1		
February				
Egypt	1			
Latvia	1	6		5
Portugal	4			4
Slovakia		1		
Spain	3			1
Total = 208	101	34	18	55
	49%	16%	9%	26%

Table 2. Antigenic analyses of influenza A H1N1 and H1N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹					
			Post infecton ferret sera					
			A/Beij 262/96	A/NC 20/99	A/Eg 96/02	A/HK 2367/04	A/Neth 128/04	A/Thess 24/05
A/Beijing/262/96			1280	320	320	640	320	640
A/New Caledonia/20/99			80	640	640	640	320	640
A/Egypt/96/02			80	320	640	640	320	320
A/Hong Kong/2637/04			80	320	320	1280	320	320
A/Netherlands/128/04			160	320	640	1280	2560	2560
A/Thessaloniki/24/05			160	320	640	1280	2560	2560
A/Switzerland/8761/2005	19.10.05	MDCKIx \1	80	640	320	NT	640	1280
A/Baden-Wurttemberg/362/2005	16.11.05	MDCKx \1	<	80	<	160	160	80
A/Lisbon/94/2005	4.12.05	MDCK1 \2	160	640	640	2560	2560	1280
A/Madagascar/2309/2005	15.12.05	MDCK1 \1	160	640	320	1280	1280	1280
A/Hong Kong/4922/2005	20.12.05	MDCK2 \1	40	320	320	640	640	320
A/Egypt/136/2005	21.12.05	E2 \1	40	320	320	NT	320	320
A/Lyon/CHU/2/06	28.12.05	MDCK2 \1	80	640	320	NT	1280	1280
A/Norway/30/2006	28.12.05	MDCK1 \1	160	640	640	NT	640	1280
A/England/488/2005	30.12.05	MDCK3 \1	40	320	320	640	640	640
A/Clermont-F/26/06	2.1.06	MDCK2 \1	160	640	640	NT	2560	2560
A/Stockholm/3/2006	4.1.06	MDCK1 \1	80	320	320	640	640	640
A/Hong Kong/35/2006	7.1.06	MDCK2 \1	160	320	640	2560	2560	1280
A/Switzerland/4243/2006	10.1.06	MDCKIx \2	80	320	640	1280	1280	1280
A/Norway/195/2006	17.1.06	MDCK1 \1	160	640	640	2560	2560	1280
A/Lisbon/13/2006	18.1.06	MDCK1 \1	160	640	320	2560	2560	2560
A/Egypt/214/2006	1.2.06	E3	160	320	320	1280	2560	2560
A/Latvia/1951/2006	11.2.06	MDCK1 \1	160	320	160	640	1280	1280
A/Barcelona/259/2006	13.2.06	MDCKx \1	160	640	320	2560	2560	2560

1. < : <40

Figure 1. Phylogenetic comparison of H1 HAs

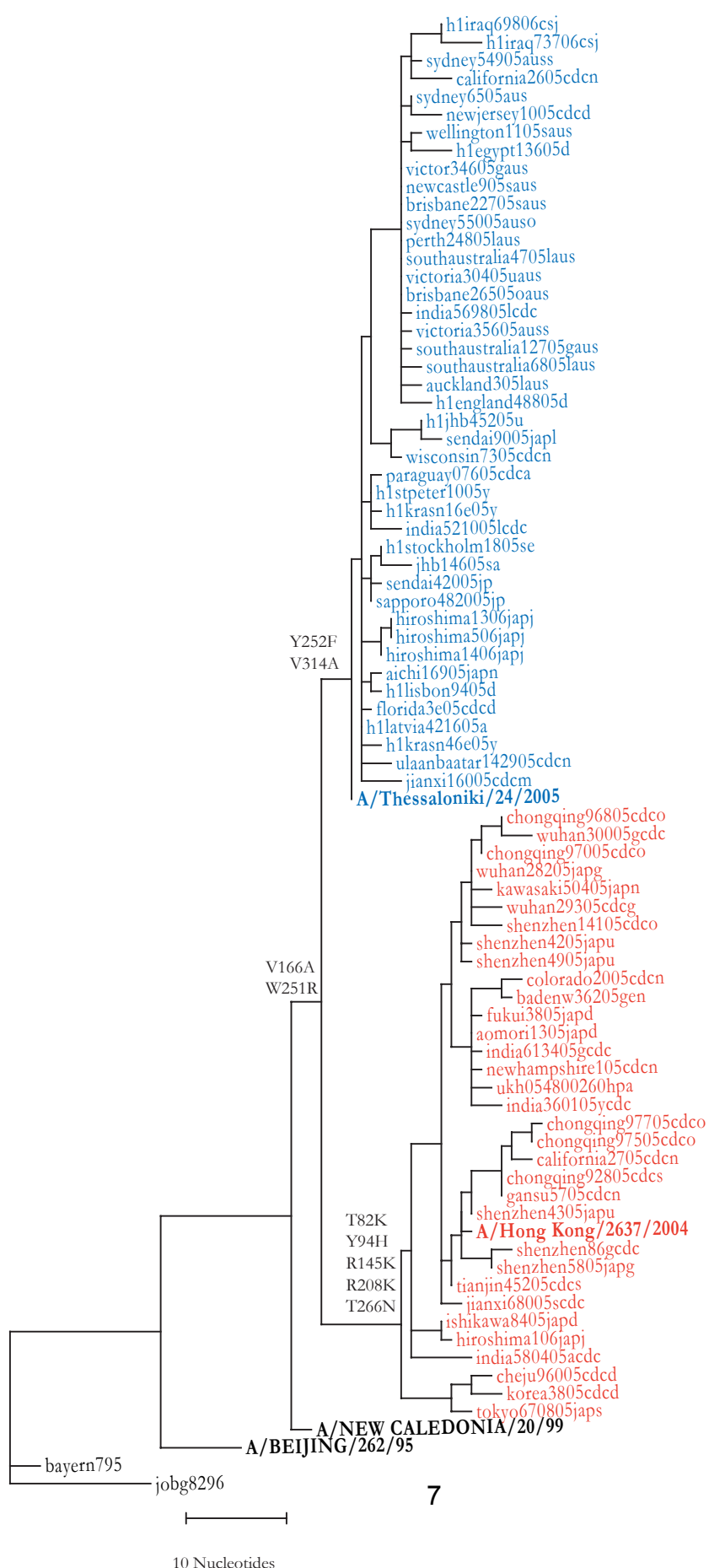


Figure 2. Phylogenetic comparison of N1 NAs

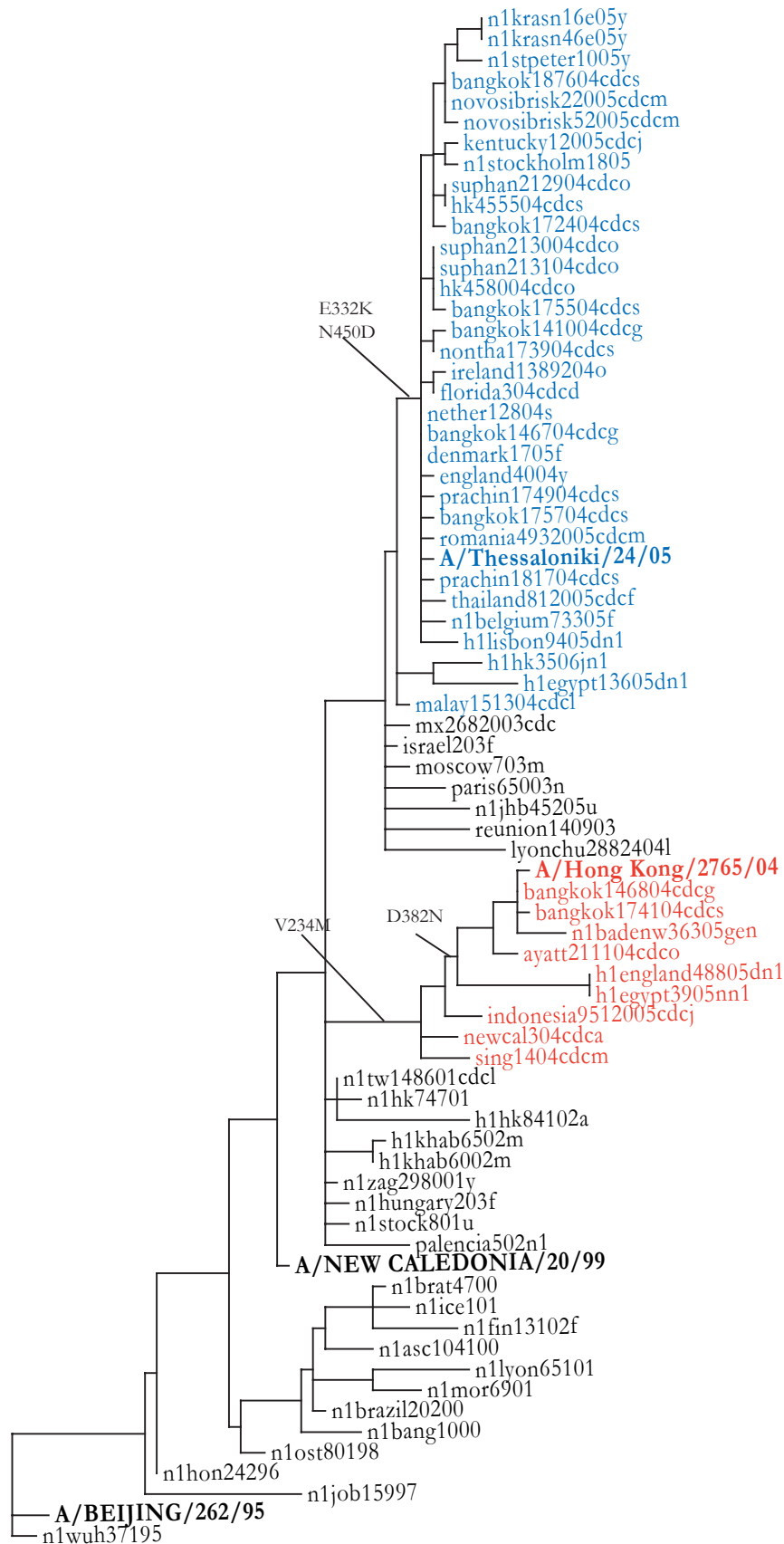


Table 3. Amino acid changes characteristic of H1N1 sequence variants

Variant Group	Representative strain	Amino acid changes ¹	
		HA	NA
1	A/Thessalonika/24/05	V166A W251R Y252F V314A	V48I E332K N450D
	A/England/488/05	V166A W251R Y252F V314A	A16T S59N I117T V234M N273S T339N D382N
2	A/HK/2637/04	T82K Y94H R145K V166A R208K W251R T266N	E214G R222Q V234M D382N

1. Relative to A/New Caledonia/20/99 (H1N1)

Table 4. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			A/Pan	X-147	A/Shan	A/Cal	A/NY	A/Ann	A/HK	A/Wis
			2007/99 F2/01	(Wy/3/03) F11/03	1219/04 F14/04	7/04 F31/05	55/04 F6/05	1138/05 F33/05	4443/05 F24/05	67/05 CDC
A/Panama/2007/1999	Ex	5120	1280	320	40	<	<	40	<	
A/Wyoming/3/2003	Ex	1280	2560	1280	640	1280	160	160	640	
A/Shantou/1219/2004	MDCKx	320	160	1280	640	1280	160	320	320	
A/California/7/2004	SpfCk3, E3	160	640	2560	2560	2560	320	640	1280	
A/New York/55/2004		<	320	1280	1280	1280	320	320	1280	
A/Annecy/1138/2005		<	40	320	160	320	320	320	160	
A/Hong Kong/4443/2005		<	80	160	160	320	320	320	160	
A/Wisconsin/67/2005		80	160	1280	640	1280	1280	2560	2560	
A/Hiroshima/52/2005	unknown	E3 \1	<	<	640	640	1280	320	640	NT
A/Umea/9/2005	10.11.05	MDCK1 \1	<	80	160	160	40	40	320	160
A/Madagascar/2102/2005	17.11.05	MDCK3 \1	<	640	1280	320	320	640	160	NT
A/Ireland/12182/05	Dec-05	MDCK2 \3	<	80	160	320	80	80	320	160
A/Hong Kong/4920/2005	19.12.05	MDCK2 \3	<	<	80	160	<	160	160	80
A/Norway/28/2006	20.12.05	MDCK1 \1	<	<	160	80	160	160	160	80
A/Stockholm/1/2006	22.12.05	MDCK1 \1	40	80	320	320	160	160	320	160
A/Netherlands/548/2005	26.12.05	MDCK1 \1	<	80	320	320	320	640	640	640
A/Denmark/112/2005	28.12.05	MDCK2 \1	<	<	<	40	40	80	80	80
A/Latvia/222/2006	4.1.06	MDCK1 \3	<	40	160	160	320	320	320	160
A/Lyon/21/06	4.1.06	MDCK2 \1	<	<	160	160	320	160	160	80
A/Finland/634/2006	6.1.06	MDCK2 \1	<	40	160	160	320	80	160	160
A/England/4/2006	9.1.06	MDCK1 \2	<	40	160	320	160	160	160	80
A/Iceland/2/2006	9.1.06	MDCK3 \1	<	<	80	160	<	160	80	80
A/Stockholm/6/2006	14.1.06	MDCK1 \1	<	40	80	160	80	160	160	160
A/Norway/143/2006	16.1.06	MDCK1 \1	<	40	<	320	320	320	320	320
A/Slovakia/134/2006	6.2.06	MDCK1 \1	<	40	160	320	160	160	320	640
A/Latvia/2047/2006	14.2.06	MDCK1 \1	<	<	80	160	160	160	160	80

1. <, <40

Figure 3. Phylogenetic comparison of H3 HAs

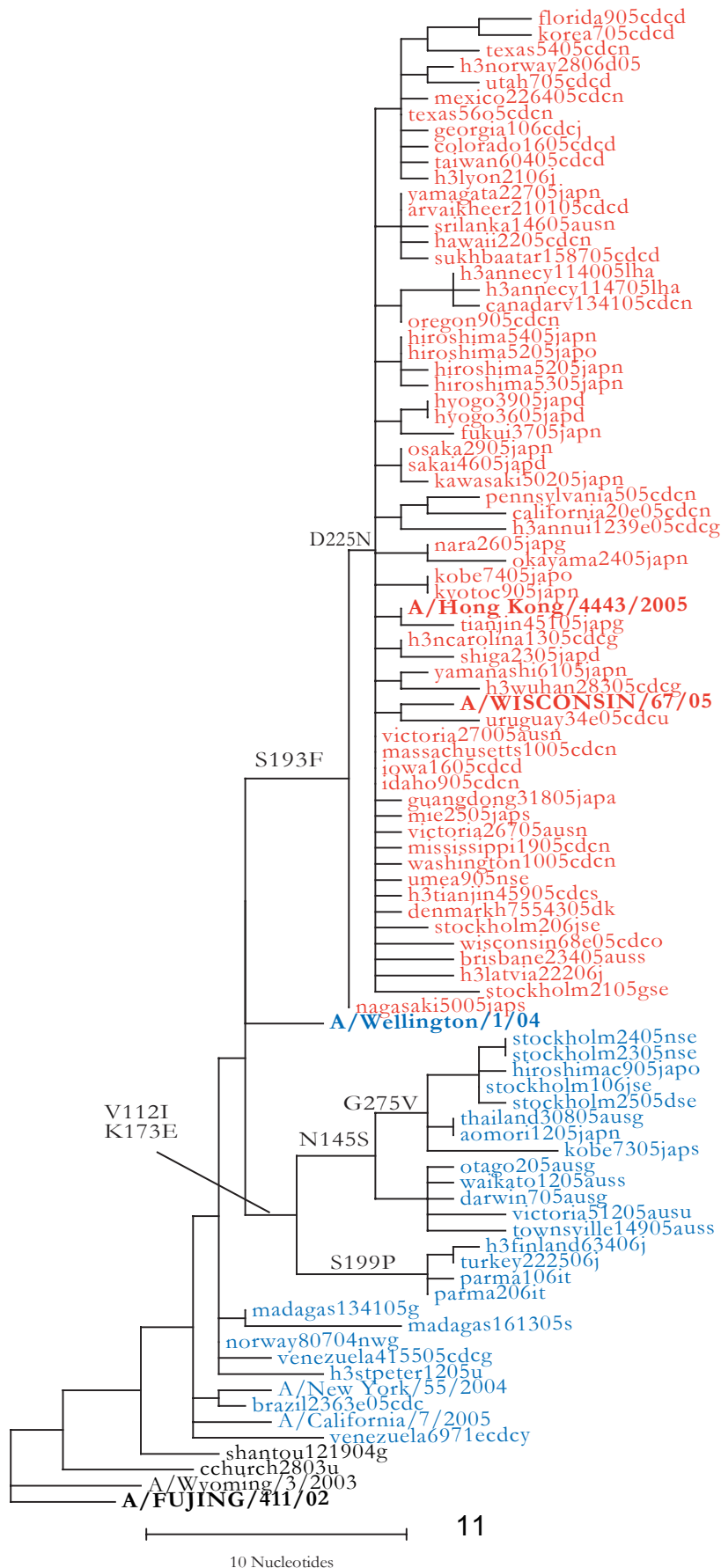


Figure 4. Phylogenetic comparison of N2 NAs

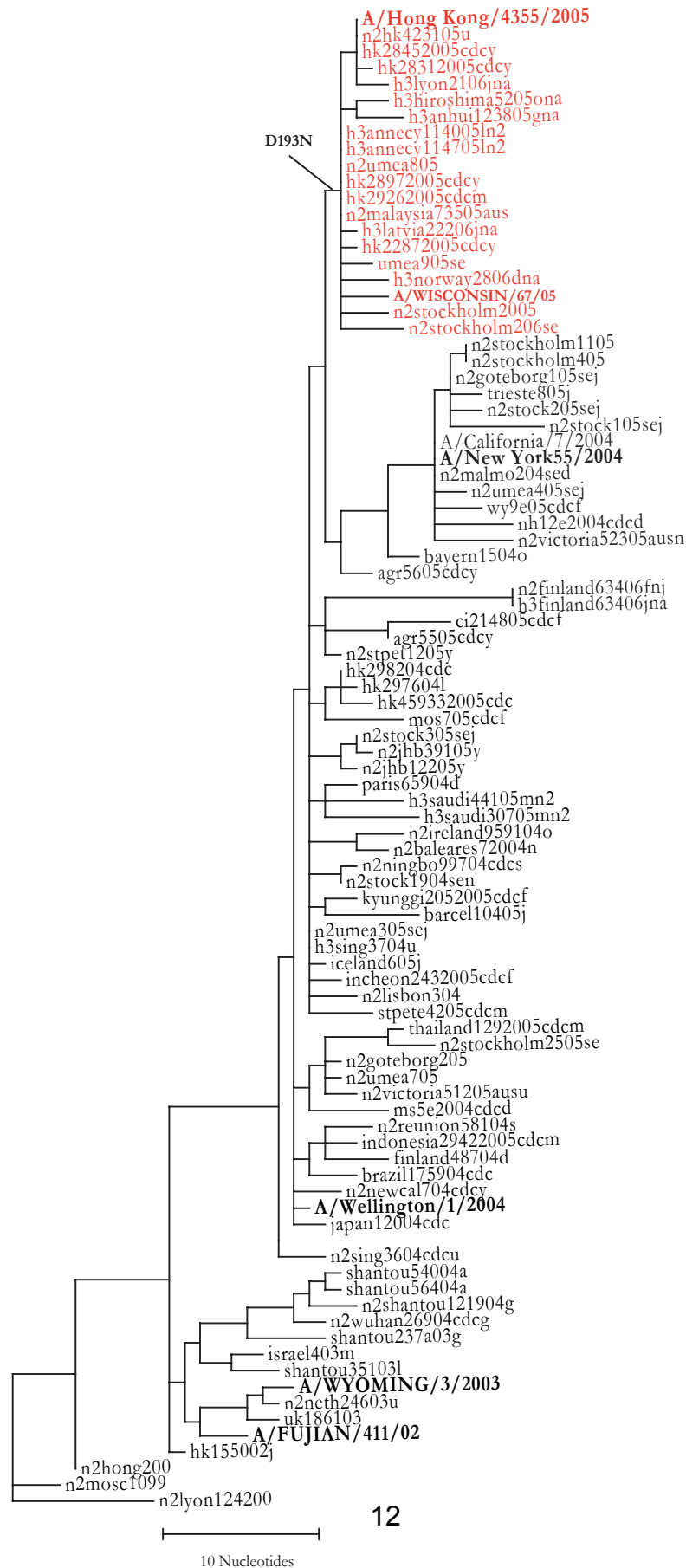


Table 5. Amino acid changes characteristic of H3N2 sequence variants

Variant group	Representative strain	Amino Acid changes ¹	
		HA	NA
1	A/Hong Kong/4443/05	S193F D225N	D93N (D151G/N)
2	A/Stockholm/23/05	V112I N145S K173E (D188Y) (G275V)	(E221K)

1. Relative to A/California/7/04.

Table 6. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Isolation date		Haemagglutination inhibition titre ¹				
			Post infection ferret sera				
			B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03	B/Eg 144/05	B/FI 7/05
B/Sichuan/379/99			160	320	<	160	160
B/Shanghai/361/2002			160	640	40	160	160
B/Jiangsu/10/2003			80	160	320	80	80
B/Egypt/144/2005			160	160	40	320	320
B/Florida/7/2005			80	160	<	320	320
B/Stockholm/2/2006	unknown	MDCK1 \1	80	160	<	160	320
B/Norway/1967/2005	7.12.05	MDCK1 \1	40	160	80	320	320
B/Norway/1991/2005	20.12.05	MDCK1 \1	40	80	<	320	160
B/Lyon/CHU/51.486/05	23.12.06	MDCK2 \1	40	160	320	160	320
B/Norway/2007/2005	29.12.05	MDCK1 \1	40	80	<	160	80
B/Lyon/CHU/1.122/06	2.1.06	MDCK2 \1	40	160	320	80	160
B/Norway/17/2006	2.1.06	MDCK1 \1	40	160	80	320	160
B/Norway/26/2006	3.1.06	MDCK1 \1	40	160	40	320	160
B/Denmark/1/2006	9.1.06	MDCK2 \1	40	40	40	320	160
B/Stockholm/2/2006	Jan-06	MDCK1 \1	80	160	<	160	320
B/Norway/174/2006	19.1.06	MDCK1 \1	80	80	320	80	320
B/Genoa/1/2006	23.1.06	MDCK1 \2	80	80	40	160	160
B/Lyon/CHU/1.122/06	2.1.06	MDCK2 \1	40	160	320	80	160
B/Denmark/1/2006	9.1.06	MDCK2 \1	40	40	40	320	160

¹ <, <40

Figure 6. Phylogenetic comparison of B HAs (Yamagata lineage)

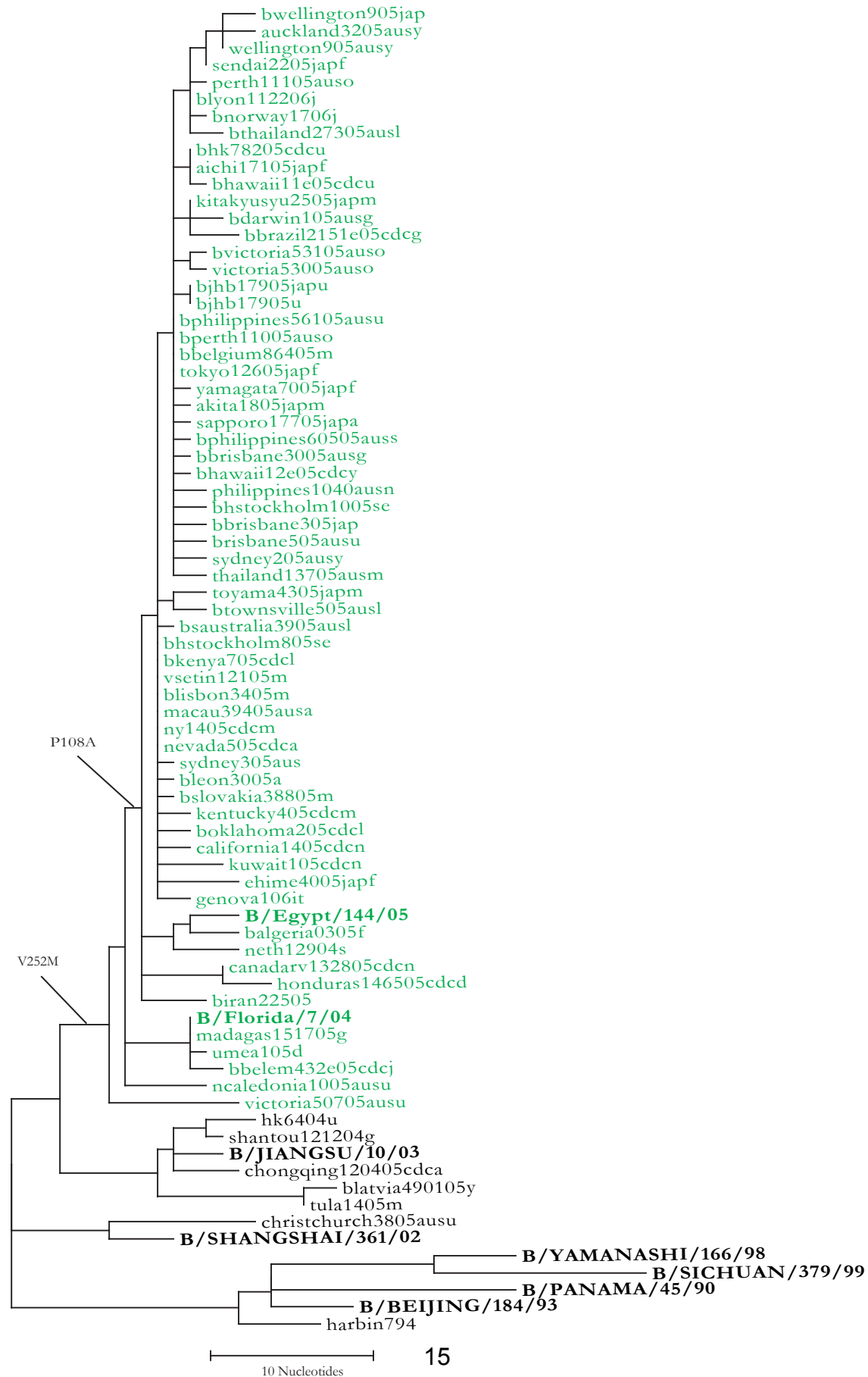


Table 7. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Isolation date		Haemagglutination inhibition titre ¹					
			B/Shan ² 7/97	Post infection ferret sera				
				B/Shan 7/97	B/Tehr 80/02	B/Bris 32/02	B/HK 45/05	B/Mal 2506/04
B/Shandong/7/97			1280	320	160	160	160	320
B/Tehran/80/2002			1280	160	160	320	160	320
B/Brisbane/32/2002			640	80	160	160	80	80
B/HK/45/2005			1280	80	80	80	160	160
B/Malaysia/2506/2004			640	160	80	160	80	320
B/England/474/2005	22.12.05	MDCK3 \1	1280	160	<	<	80	80
B/Lyon/CHU/52.133/05	26.12.05	MDCK2 \1	2560	160	<	<	160	80
B/Toulouse/43/06	13.12.05	MDCK2 \1	1280	160	<	<	160	40
B/Norway/1992/2005	20.12.05	MDCK1 \1	1280	160	<	<	160	80
B/Toulouse/44/06	21.12.05	MDCK2 \1	2560	160	<	<	160	40
B/Hong Kong/15/2006	30.12.05	MDCK2 \1	1280	320	80	40	160	160
B/Switzerland/4081/2006	5.1.06	MDCKIx \1	1280	160	<	<	160	80
B/Stockholm/1/2006	11.1.06	MDCK0 \1	1280	160	40	40	80	160
B/England/465/2005	2.12.05	MDCK3 \1	1280	320	40	40	160	80
B/Egypt/208/2006	25.1.06	C1 E1	1280	80	40	<	80	160
B/Ireland/2193/2006	Jan-06	MDCKx \1	1280	160	<	<	160	80
B/Genoa/2/2006	30.1.06	MDCK1 \1	1280	160	<	<	320	160
B/Netherlands/002/2006	19.1.06	MDCK1 \1	2560	160	<	<	160	80
B/Norway/196/2006	24.1.06	MDCK1 \1	1280	160	<	<	160	80
B/Slovakia/117/2006	30.1.06	MDCK1 \1	1280	40	<	<	160	80
B/Latvia/1844/2006	9.2.06	MDCK1 \1	1280	40	<	<	160	80
B/Lisbon/22/2006	4.2.06	MDCK1 \1	1280	40	<	<	160	80
B/Barcelona/258/2006	10.2.06	MDCKx \1	1280	80	<	<	160	80

¹ <, <40

² hyperimmune sheep serum

Figure 7. Phylogenetic comparison of B HAs (Victoria lineage)

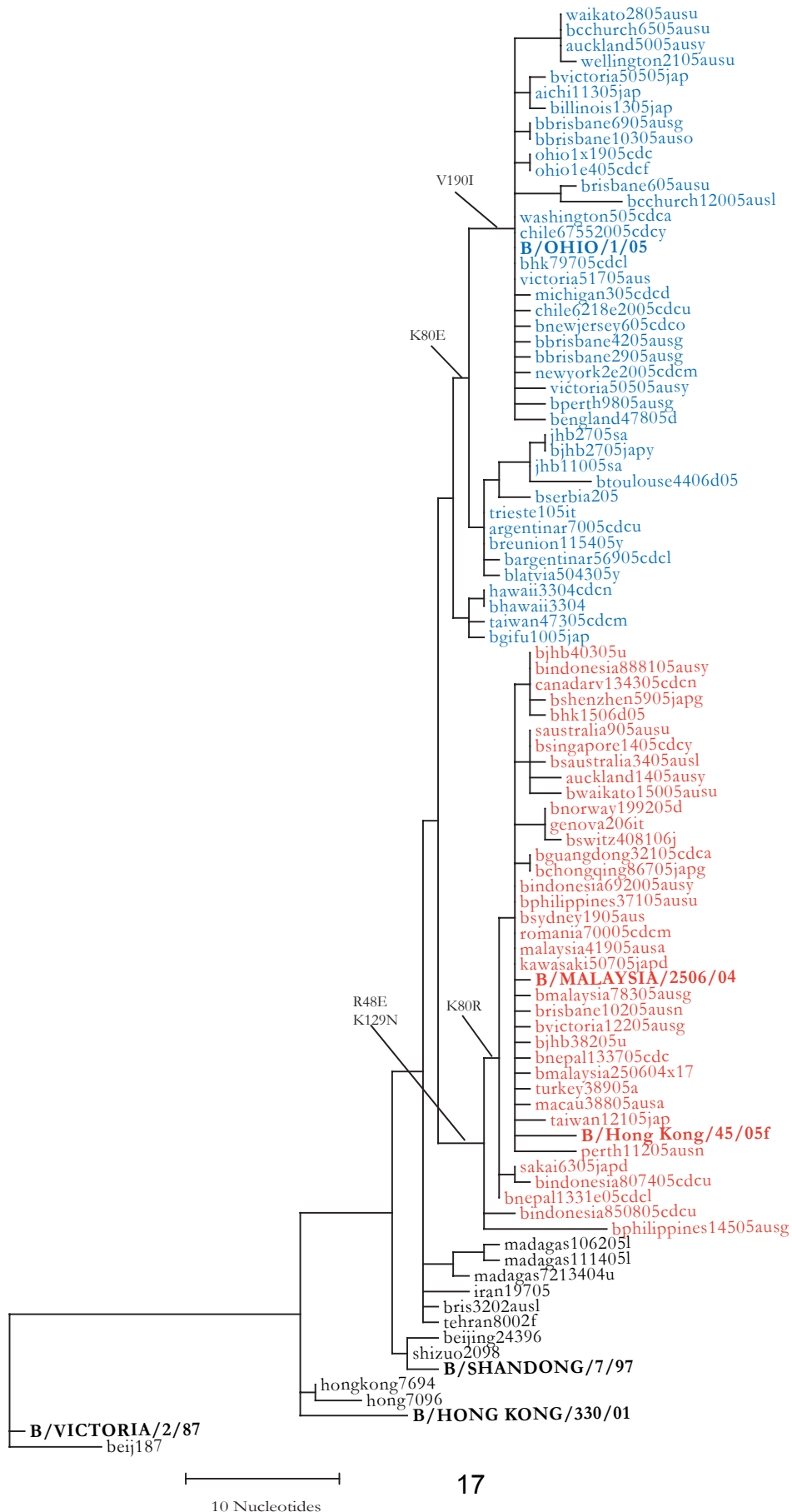


Table 8. Amino acid changes characteristic of recent B sequence variants

HA Lineage	Representative strain	Amino acid changes	
		HA	NA
Yamagata ¹	B/Egypt/144/05	V252M	T42Q T49I K186R N219K N235D S244P D329N N342D D392E E436T
	B/Belgium/864/05	V252M P108A	
Victoria ²	B/Malaysia/2506/04	R48E K80R K129N	S50P P51S I248V V271I L396F D463N
	B/Ohio/1/05	K80E V190I	K125N I248V K373E E404G/K

1. Relative to B/Shanghai/361/02. 2. HA relative to B/Shandong/7/97; NA relative to B/Tehran/80/02.

Figure 8. Phylogenetic comparison of B NAs

